

Billy Tsouvalas

PhD, MSc, BSc, Research Assistant at SUNY Stony Brook
IBM & Google Cybersecurity Professional Certifications
[Website](#), [LinkedIn](#), [Google Scholar](#)

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Education

Stony Brook University	2022-2026
PhD Computer Science	
Dissertation: Securing Web Applications with Deception-based Defenses	
Stony Brook University	2022-2023
MSc Computer Science	
Virginia Commonwealth University	2021-2022
Computer Science Graduate Studies	
University of Patras	2014-2020
BSc, MSc in Electrical and Computer Engineering	
Master's Thesis: Malware Classification Methodologies	

Work Experience

Cloudflare	Summer 2024, Summer 2025
Research Scientist Intern	
Bot & Fraud Research and Development	
Research Assistant	2023-Present
Stony Brook University, Stony Brook, NY	
Web Security, Deception-based Defenses	
Bot Management and Mitigation	
System Administrator	2023-Present
Stony Brook University, Stony Brook, NY	
Computer Science Dept PragSec Lab Network & Server administrator	
Teaching Assistant	2022-2023
Stony Brook University, NY	
Computer Networks course	
Research Assistant	2021-2022
Virginia Commonwealth University, Richmond, VA	
Network Intrusion Detection and Prevention Systems	
Critical Infrastructure Cybersecurity: Energy Sector (Solar, Wind)	
Publicity and Media Administrator	2021-2022
Virginia Commonwealth University Cybersecurity Center, Richmond, VA	
Publicity, Social Media Marketing, and Digital Presence	
Research Intern	2019-2020
University of Patras, Greece	
Static and Dynamic Malware Analysis and Detection	
Online Misinformation Propagation	
Freelance Web Developer	2018-2020
Web Design, Full-Stack Web Development	
Business-oriented Web Applications	
Social Media and Digital Marketing Strategy	

Research

SBU PragSec Lab	2022-Present
Web Application Security, Honeypots, Bot behavior Active Defenses and Deception-based Security	
VCU Modern Heuristics Research Group	2021-2022
Critical Infrastructure and Network Security, Cyber-Physical Systems Anomaly Detection for IDS (Collaboration with DoE)	
University of Patras Malware Analysis Team	2019-2020
Development of Static and Dynamic Malware Detection Tools Online Rumor Propagation and Misinformation Detection Framework	

Publications

	2025
Defense of the Clones: Securing Web Applications with Automatic Honeypot Generation and Deployment <i>Billy Tsouvalas, Nick Nikiforakis</i> 2025 APWG Symposium on Electronic Crime Research (eCrime)	
[PATENT] Methods and systems for surgical training <i>Lauren Siff, Milos Manic, Lewis Franklin Bost, Vasileios Tsouvalas</i> Patent number: US20250239177A1, Application number: 18/853,112	2024
Knocking on Admin's Door: Protecting Critical Web Applications with Deception <i>Billy Tsouvalas, Nick Nikiforakis</i> International Conference on Detection of Intrusions and Malware, and Vulnerability Assessment (DIMVA 2024)	2023
Efficient Graph-Based Malware Detection Using Minimized Kernel and SVM <i>Billy Tsouvalas, Dimitrios Serpanos</i> Cyber Malware: Offensive and Defensive Systems	
Surgical teaching of the retropubic midurethral sling: a virtual reality training system <i>Lauren Siff, Vasileios Tsouvalas, Lewis F Bost, Milos Manic</i> International Urogynecology Journal - AWARD: Best Video Presentation at AUGS/IUGA	2022
Location and Language Independent Fake Rumor Detection through Epidemiological and Structural Graph Analysis of Social Connections <i>Dimitrios Serpanos, Georgios Xenos, Billy Tsouvalas</i> Frontiers in Artificial Intelligence	2021
Data-driven correlation of cyber and physical anomalies for holistic system health monitoring <i>Daniel L Marino, Chathurika S Wickramasinghe, Billy Tsouvalas, Craig Rieger, Milos Manic</i> IEEE Access	
CyStAR: Cyber-Physical Threat Detection and Asset Monitoring <i>Craig Rieger, Jacob P Lehmer, Daniel Marino Lizarazo, Chathurika S Wickramasinghe, Billy Tsouvalas, Milos Manic</i> Idaho National Laboratory Software	
Cyber State Awareness For Resilience <i>Craig Rieger, Daniel Marino Lizarazo, Chathurika S Wickramasinghe, Billy Tsouvalas, Milos Manic, Jacob</i>	

P Lehmer
Idaho National Laboratory Software

Academic Service - Volunteering

Active Defence and Deception Workshop (AD&D) 2026	2025
PC Member	
Annual Computer Security Applications Conference (ACSAC) 2025	2025
Reviewer	
USENIX 2025	2025
Artifact Evaluation Committee	
Information Security Conference (ISC) 2024	2024
Web Security Session Chair, Publicity Chair	
USENIX WOOT Conference on Offensive Technologies 2024	2024
Artifact Evaluation Committee	
ACM CCS 2024	2024
Artifact Evaluation Committee	
USENIX 2024	2024
Artifact Evaluation Committee	
<i>Women in Science and Engineering (WSE)</i>	Spring 2024
Instructor for SBU Course on Web Security and Network Intrusion Course	
<i>Women in Science and Engineering (WSE)</i>	Fall 2023
Instructor for SBU Course on Web Security and Network Intrusion Course	
Publicity Chair IEEE EuroS&P Workshop	2022, 2023
Active Defense & Deception Workshop	
IEEE Transactions on Industrial Informatics	2022
Reviewer	

Awards - Grants

Distinguished Artifact Reviewer ACM CCS 2024	2024
Best Video Abstract Presentation at AUGS/IUGA 2022	2022
Virtual Reality Training Model for Retropubic Sling Surgery American Urogynecologic Society (AUGS) and International Urogynecological Association (IUGA) 2022	
NSF Student Travel Grant for 2020 IEEE/IFIP Network Operations and Management Symposium (NOMS'2020)	2021
Award Abstract # 2014439	

Invited Talks

Center of Academic Excellence in Cyber Research (CAE-R)	2025
Knocking on Admin's Door: Protecting Critical Web Applications with Deception	

Certifications & Courses

INFOSEC OWASP Top 10 Specialization Course duration: 1 month	Obtained: Sep. 2023
Google Cybersecurity Professional Certificate Course duration: 6 months	Obtained: Aug. 2023
Palo Alto Networks Cybersecurity Professional Certificate Course duration: 1 month	Obtained: June 2023
CISCO Network Automation Engineering Fundamentals Specialization Course duration: 3 months	Obtained: June 2023
(ISC) ² Certified in Cybersecurity Specialization (CC) Course duration: 1 month	Obtained: June 2023
EC-Council Cybersecurity Attack and Defense Fundamentals Course duration: 5 months	Obtained: June 2023
INFOSEC Cyber Incident Response Specialization Course duration: 4 months	Obtained: Aug. 2021
IBM Cybersecurity Analyst Professional Certificate Course duration: 8 months	Obtained: Dec. 2020
Homeland Security and Cybersecurity Specialization University of Colorado, Colorado Springs Course duration: 4 months	Obtained: Dec. 2020

Full list available on [Website](#).

Professional Skills

Network & Systems Nginx, Apache, Mitmproxy, Openresty, Lua, Docker, Hydra, Wpscan, Honeypot Deployment, Bot Behavior Analysis	Network Security IDS/IPS, Wireshark, Zeek, Suricata, Snort, Stratosphere IPS, Anomaly Detection, Intrusion Detection Datasets
Penetration Testing and Forensics OWASP, Burp Suite, Metasploit, NMAP, Hashcat, Aircrack-ng, Volatility	Malware Analysis NSA Ghidra, Hex Rays IDA Pro, Radare, gdb, Malware Reverse Engineering, Control Flow & API Call Graphs
Machine Learning and Data Mining Binary Decision Trees, Random Forest, SVM, Neural Networks, Markov Chains, BERT	Programming and Scripting Python, PHP, Rust, C, C++, DSP Assembly, Java, Javascript, Git, Matlab

Languages

English ¹	● ● ● ● ●
French ²	● ● ● ● ●
Greek (Native)	● ● ● ● ●
German	● ● ● ● ●

¹ETS TOEFL
²DALF C1 CERF

Soft Skills

- Fast Learner
- Responsible
- Performance Under Pressure
- Team Work
- Multitasking
- Stamina
- Verbal Communication
- Public Speaking

Misc

Military Service Obligations Fulfilled 2018-2019
Hellenic Navy